



AROR UNIVERSITY OF ART, ARCHITECTURE, DESIGN & HERITAGE, SUKKUR, SINDH

Course: Applied Physics

Lecture 05

Energy and Work

- Energy is the ability to do work.
- Work is the energy transferred to or from a system by a force that acts on it.



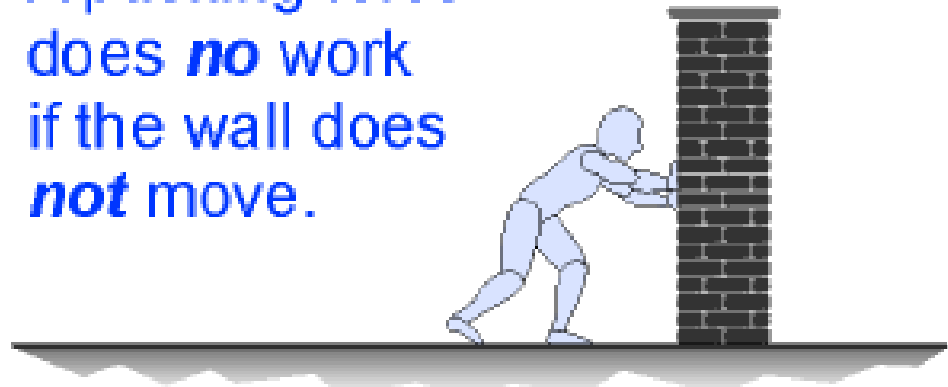
Mechanical

- **Mechanical energy** is the energy which is possessed by an object due to its motion or its stored energy of position
 - **Kinetic energy** : is the energy of motion
 - **Potential Energy** : an object can store energy as the result of its position or elastic source

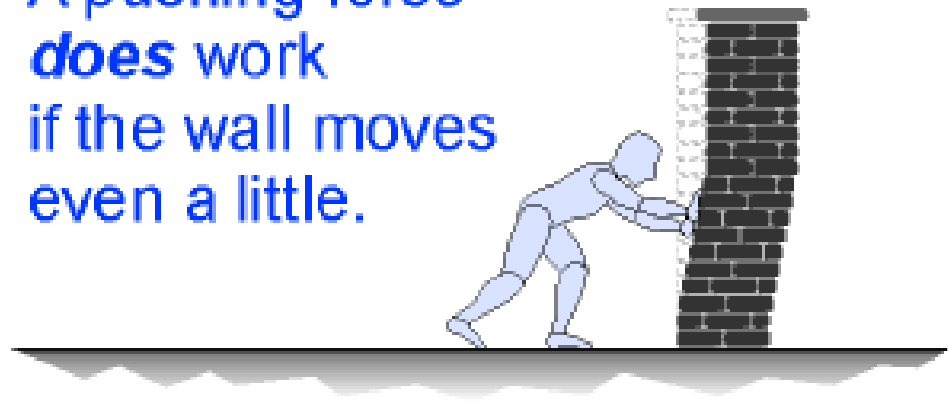
Work

- In physics, **work** has a very specific meaning.
- In physics, work represents a measurable change in a system, caused by a force.

A pushing force does **no** work if the wall does **not** move.



A pushing force **does** work if the wall moves even a little.



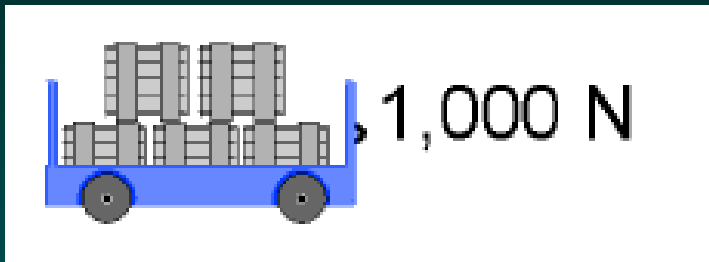


Work Concept

- **Work** is defined as a force acting upon an object to cause a displacement
- Mathematically, work can be expressed by the following equation.
- $W = F \times d \cos \theta$ ($\cos 0^\circ = 1$)
- where F = force, d = displacement, and the angle (θ) is defined as the angle between the force and the displacement vector



Work (force is parallel to distance)



Work (joules) $\longrightarrow$ $W = F \times d$

Force (N) $\swarrow$

Distance (m) $\swarrow$

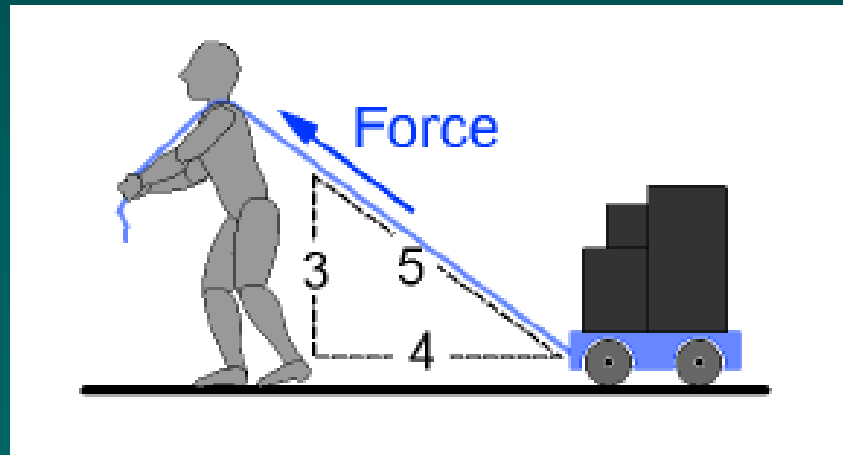


Work (force at angle to distance)

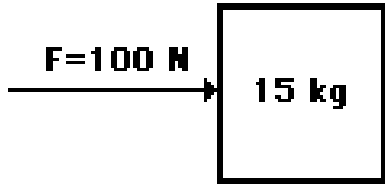
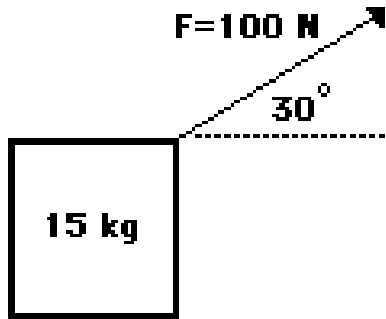
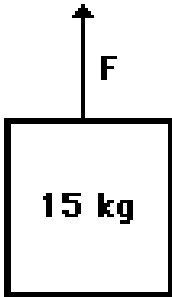
Work (joules) → $W = Fd \cos (q)$ ← **Angle**

Force (N)

Distance (m)



Work Calculations

Diagram A	Diagram B	Diagram C
		
A 100 N force is applied to move a 15 kg object a horizontal distance of 5 meters at constant speed.	A 100 N force is applied at an angle of 30° to the horizontal to move a 15 kg object at a constant speed for a horizontal distance of 5 m.	An upward force is applied to lift a 15 kg object to a height of 5 meters at constant speed.

$$W = F \times d$$

$$= 100\text{ N} \times 5\text{ m}$$

$$= 500\text{ N m}$$

$$W = F \times d \cos 30^\circ$$

$$= 100\text{ N} \times 5\text{ m} \times .87$$

$$= 413\text{ N m}$$

$$W = F \times d$$

$$= 15\text{ Kg}(10\text{ m/s}^2) \times 5\text{ m}$$

$$= 750\text{ N m}$$

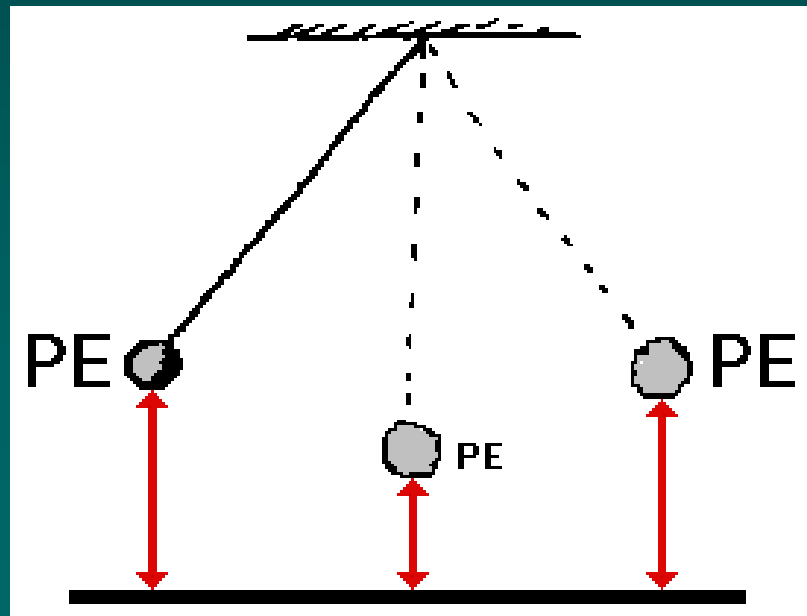
Gravitational Potential Energy

- After an object has been lifted to a height, work is done.



- $PE = W = F \times d = mgh$

Potential Energy is maximum at the maximum HEIGHT





Potential Energy Calculation

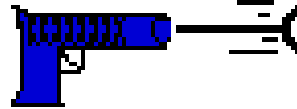
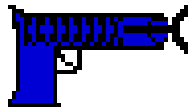
- How much potential energy is lost by a 5Kg object to kinetic energy due a decrease in height of 4.5 m
- $PE = mgh$
- $PE = (5\text{Kg})(10 \text{ m/s}^2)(4.5 \text{ m})$
- $PE = 225 \text{ Kg m}^2/\text{s}^2$
- $PE = 225 \text{ J}$



Kinetic Energy Calculation

- The energy of motion
- $\Delta KE = W = F \times d = mgh = \frac{1}{2} mv^2$
- Find the kinetic energy of an 4 Kg object moving at 5m/s.
- $KE = \frac{1}{2} mv^2$
- $KE = \frac{1}{2} (4\text{Kg})(5\text{m/s})^2$
- $KE = 50 \text{ Kg m}^2 / \text{s}^2$
- $KE = 50 \text{ J}$

Elastic potential energy



The compressed springs of a dart gun store elastic potential energy. When the trigger is pulled, the springs apply a force to do work on the dart.

$$F_{\text{spring}} = k \times x$$

$$PE_{\text{spring}} = \frac{1}{2} \times k \times x^2$$

where k = spring constant

x = amount of compression

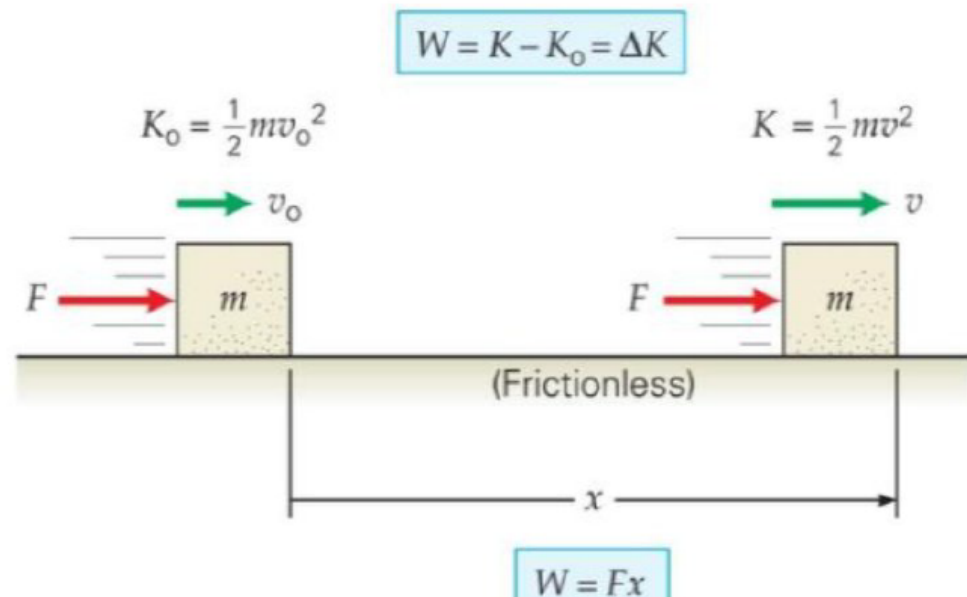
(relative to equilibrium pos'n)

The Work-Energy Theorem: Kinetic Energy

We'll start with the form of energy that is most closely related to work – **kinetic energy (KE)**.

Consider the figure below, in which an object of mass m has an initial speed v_0 . A constant force F acts on the object (in the same direction as its velocity), increasing the speed to v . The displacement of the object is x .

From the last lecture, we know that the work done on the object is $W = Fx$. Then, from chapter 2, we have $v^2 = v_0^2 + 2ax$, which implies that the object has acceleration $a = \frac{v^2 - v_0^2}{2x}$.



Work-Kinetic Energy Theorem: Kinetic Energy

Now, we apply Newton's second law:

$$F = ma = m \left(\frac{v^2 - v_0^2}{2x} \right)$$

Finally, the work done is

$$W = Fx = m \left(\frac{v^2 - v_0^2}{2x} \right) x = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2.$$

We now define the **kinetic energy (K)** of the object:

$$K = \frac{1}{2}mv^2$$

By this definition, the work done on the object is simply equal to the change in the object's kinetic energy:

$$W = K - K_0 = \Delta K$$

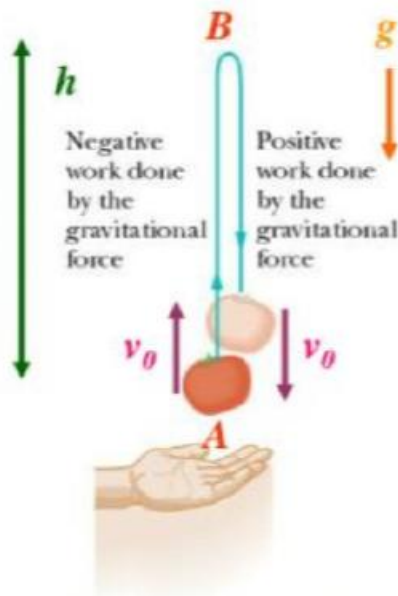
Work-Kinetic Energy Theorem: Kinetic Energy

Kinetic energy can be thought of as **energy possessed by mass that has motion**. It has SI units of Joules, and can not have a negative value (since both m and v^2 are either positive or zero). The last equation on the previous slide represents the **work-kinetic energy theorem**:

the net work done on a body by all the forces acting on it is equal to the change in kinetic energy of the body

When positive net work is done on an object, its kinetic energy increases (which means its speed increases, assuming that mass is constant). When negative net work is done, the object's kinetic energy and speed must decrease. If net work is zero, the object's KE and speed do not change.

Work and Potential Energy:

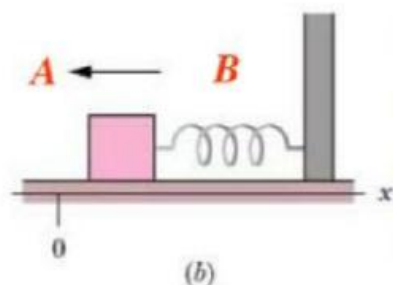
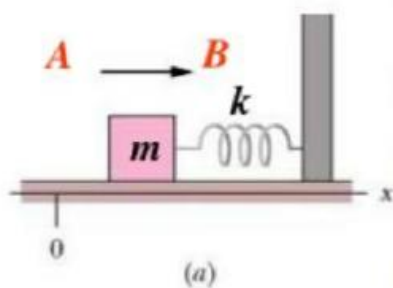


Consider the tomato of mass m shown in the figure. The tomato is taken together with the Earth as the system we wish to study. The tomato is thrown upward with initial speed v_0 at point A. Under the action of the gravitational force it slows down and stops completely at point B. Then the tomato falls back and by the time it reaches point A its speed has reached the original value v_0 .

Below we analyze in detail what happens to the tomato-Earth system. During the trip from A to B the gravitational force F_g does negative work $W_1 = -mgh$. Energy is transferred by F_g **from** the kinetic energy of the tomato **to** the gravitational potential energy U of the tomato-Earth system. During the trip from B to A the transfer is reversed. The work W_2 done by F_g is positive ($W_2 = mgh$). The gravitational force transfers energy **from** the gravitational potential energy U of the tomato-Earth system **to** the kinetic energy of the tomato. The change in the potential energy U is defined as

$$\Delta U = -W$$

(8-2)



Consider the mass m attached to a spring of spring constant k as shown in the figure. The mass is taken together with the spring as the system we wish to study. The mass is given an initial speed v_0 at point A. Under the action of the spring force it slows down and stops completely at point B, which corresponds to a spring compression x . Then the mass reverses the direction of its motion and by the time it reaches point A its speed has reached the original value v_0 .

As in the previous example, we analyze in detail what happens to the mass-spring system. During the trip from A to B the spring force F_s does negative work $W_1 = -kx^2/2$. Energy is transferred by F_s **from** the kinetic energy of the mass **to** the potential energy U of the mass-spring system.

During the trip from B to A the transfer is reversed. The work W_2 done by F_s is positive ($W_2 = kx^2/2$). The spring force transfers energy **from** the potential energy U of the mass-spring system **to** the kinetic energy of the mass. The change in the potential energy U is defined as

$$\Delta U = -W$$

(8-3)

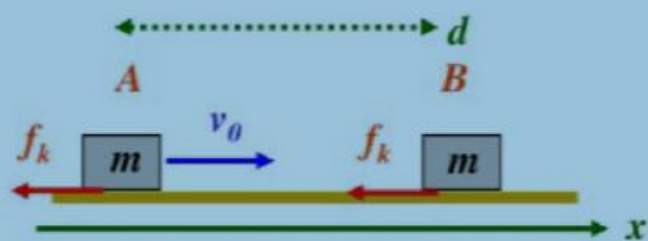
Conservative and Nonconservative Forces.

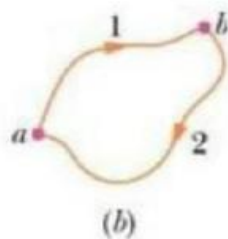
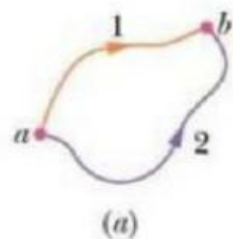
The gravitational force and the spring force are called “**conservative**” because they can transfer energy from the kinetic energy of part of the system to potential energy and vice versa.

Frictional and drag forces on the other hand are called “**nonconservative**” for reasons that are explained below.

Consider a system that consists of a block of mass m and the floor on which it rests. The block starts to move on a horizontal floor with initial speed v_0 at point A. The coefficient of kinetic friction between the floor and the block is μ_k . The block will slow down by the kinetic friction f_k and will stop at point B after it has traveled a distance d . During the trip from point A to point B the frictional force has done work $W_f = -\mu_k mgd$. The frictional force transfers energy from the kinetic energy of the block to a type of energy called **thermal** energy. This energy transfer **cannot be reversed**. Thermal energy cannot be transferred back to kinetic energy of the block by the kinetic friction. This is the hallmark of nonconservative forces.

(8-4)





Path Independence of Conservative Forces

In this section we will give a test that will help us decide whether a force is conservative or nonconservative.

A force is conservative if the net work done on a particle during a round trip is always equal to zero (see fig. b).

$$W_{\text{net}} = 0$$

Such a round trip along a closed path is shown in fig.b. In the examples of the tomato-Earth and mass-spring system $W_{\text{net}} = W_{ab,1} + W_{ba,2} = 0$.

We shall prove that if a force is conservative then the work done on a particle between two points a and b does **not** depend on the path.

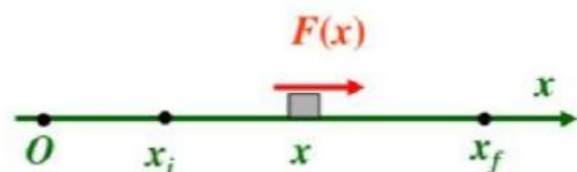
From fig. b we have: $W_{\text{net}} = W_{ab,1} + W_{ba,2} = 0 \rightarrow W_{ab,1} = -W_{ba,2}$ (eq. 1)

From fig.a we have $W_{ab,2} = -W_{ba,1}$ (eq. 2)

If we compare eq. 1 and eq. 2 we get:

$$W_{ab,1} = W_{ab,2}$$

(8-5)



Determining Potential Energy Values :

In this section we will discuss a method that can be used to determine the difference in potential energy ΔU of a conservative force F between points x_f and x_i on the x -axis if we know $F(x)$.

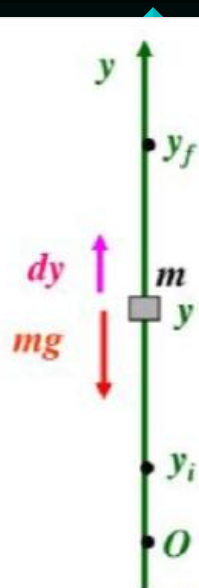
A conservative force F moves an object along the x -axis from an initial point x_i to a final point x_f . The work W that the force F does on the object is given by

$W = \int_{x_i}^{x_f} F(x)dx$. The corresponding change in potential energy ΔU was defined as

$\Delta U = -W$. Therefore the expression for ΔU becomes

$$\Delta U = - \int_{x_i}^{x_f} F(x)dx$$

(8-6)



Gravitational Potential Energy :

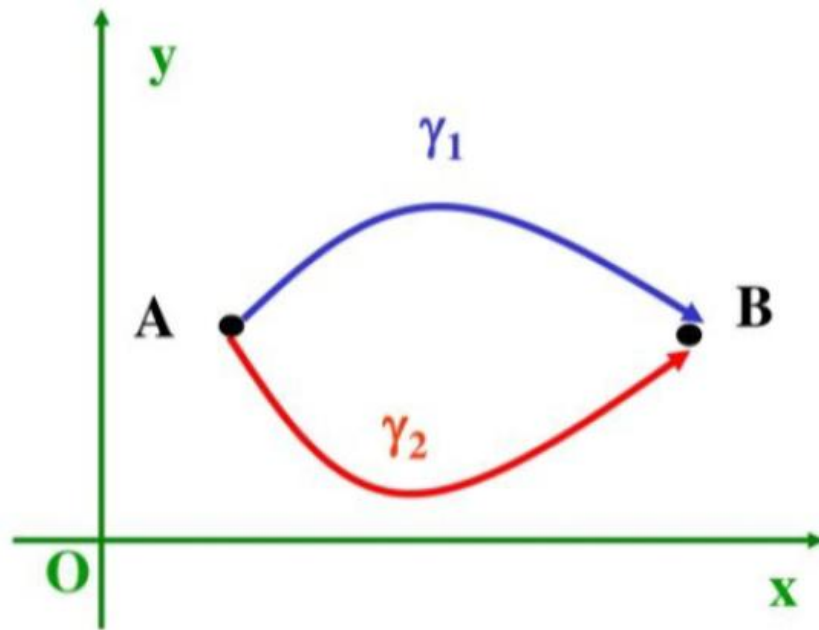
Consider a particle of mass m moving vertically along the y -axis from point y_i to point y_f . At the same time the gravitational force does work W on the particle, which changes the potential energy of the particle-Earth system. We use the result of the previous section to calculate ΔU .

$$\Delta U = - \int_{y_i}^{y_f} F(x) dy \quad F = -mg \quad \rightarrow \quad \Delta U = - \int_{y_i}^{y_f} (-mg) dy = mg \int_{y_i}^{y_f} dy = mg [y]_{y_i}^{y_f}$$

$\Delta U = mg(y_f - y_i) = mg\Delta y$. We assign the final point y_f to be the "generic" point y on the y -axis whose potential is $U(y) \rightarrow U(y) - U_i = mg(y - y_i)$. Since only changes in the potential are physically meaningful, this allows us to define arbitrarily y_i and U_i . The most convenient choice is $y_i = 0$, $U_i = 0$. This particular choice gives

$$U(y) = mgy$$

(8-7)



Potential Energy U

U is defined for conservative forces **only**

Point A (x_A, y_A)

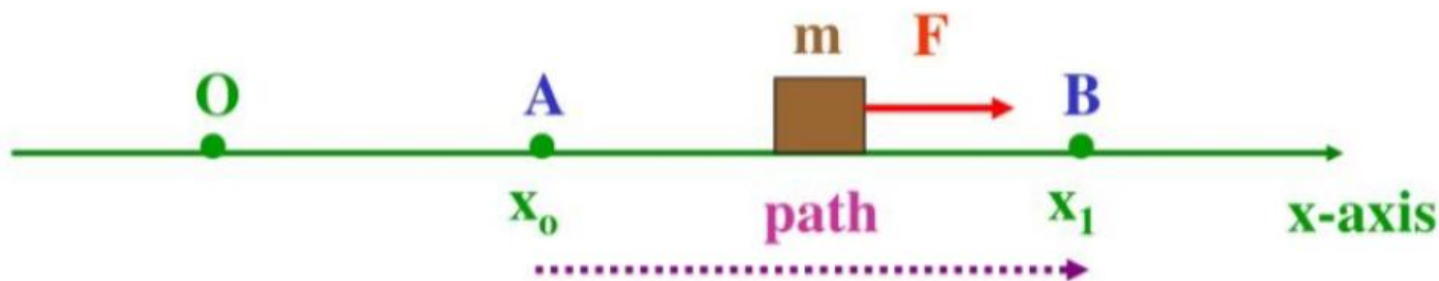
Point B (x_B, y_B)

W_{AB} = work done by a conservative force $\mathbf{F}$ from A to B

W_{AB} does **not** depend on the path taken. This can be expressed mathematically as follows: $W_{AB} = W(A, B)$

Also it can be written as: $W_{AB} = U(A) - U(B)$

U is the potential energy. **Units: J**

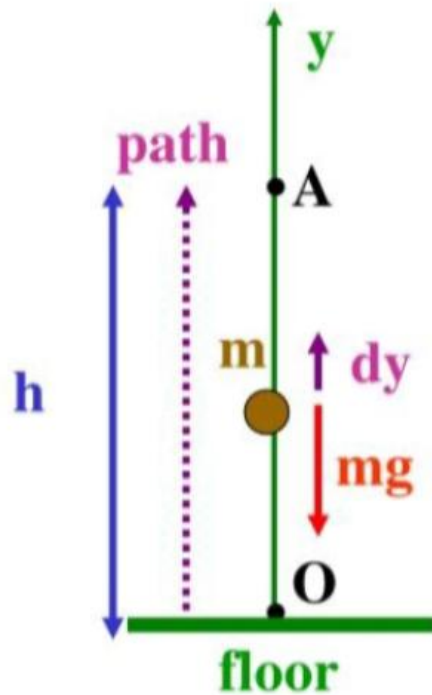


We assume that F depends on position i.e. $F = F(x)$

$$W_{AB} = U(x_0) - U(x_1) \quad \text{Also} \quad W_{AB} = \int_{x_0}^{x_1} F(x) dx \quad \rightarrow$$

$$U(x_1) - U(x_0) = - \int_{x_0}^{x_1} F(x) dx$$

Point A at x_0 is called the “**reference point**” . The choice of x_0 and $U(x_0)$ does not affect the result of our calculations. Thus we will choose x_0 and $U(x_0)$ for maximum convenience



Example: Potential energy of the gravitational force

Reference point: Point O at $y_0 = 0$

Point A at $y_1 = h$

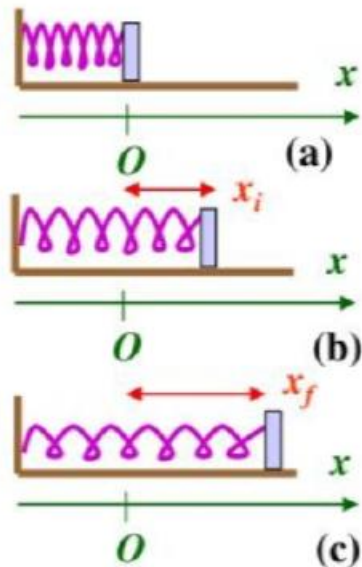
$$U(y_1) - U(y_0) = -\int_0^h F_y dy = -\int_0^h (-mg) dy = mg \int_0^h dy$$

$$U(y_1) - U(y_0) = mg [y]_0^h = mgh$$

We choose: $U(y_0) = 0$

Thus we get :

$$U = mgh$$



Potential Energy of a Spring :

Consider the block-mass system shown in the figure.

The block moves from point x_i to point x_f . At the same time the spring force does work W on the block, which changes the potential energy of the block-spring system by an amount

$$W = \int_{x_i}^{x_f} F(x)dx = \int_{x_i}^{x_f} -kx dx = -k \int_{x_i}^{x_f} x dx. \quad \Delta U = -W \rightarrow$$

$$\Delta U = k \left[\frac{x^2}{2} \right]_{x_i}^{x_f} = \frac{kx_f^2}{2} - \frac{kx_i^2}{2}. \quad \text{We assign the final point } x_f \text{ to be the "generic"}$$

point x on the x -axis whose potential is $U(x) \rightarrow U(x) - U_i = \frac{kx^2}{2} - \frac{kx_i^2}{2}.$

Since only changes in the potential are physically meaningful, this allows us to define arbitrarily x_i and U_i . The most convenient choice is $x_i = 0$, $U_i = 0$. This particular choice gives

$$U = \frac{kx^2}{2} \quad (8-8)$$

Conservation of Mechanical Energy :

Mechanical energy of a system is defined as the sum of potential and kinetic energies $E_{\text{mech}} = K + U$. We assume that the system is isolated, i.e., no external forces change the energy of the system. We also assume that all the forces in the system are conservative. When an internal force does work W on an object of the system this changes the kinetic energy by $\Delta K = W$ (eq. 1). This amount of work also changes the potential energy of the system by an amount $\Delta U = -W$ (eq. 2).

If we compare equations 1 and 2 we have: $\Delta K = -\Delta U \rightarrow$

$K_2 - K_1 = -(U_2 - U_1) \rightarrow K_1 + U_1 = K_2 + U_2$. This equation is known as the **principle of conservation of mechanical energy**. It can be summarized as

$$\Delta E_{\text{mech}} = \Delta K + \Delta U = 0$$

For an isolated system in which the forces are a mixture of conservative and nonconservative forces the principle takes the following form: $\Delta E_{\text{mech}} = W_{nc}$

Here, W_{nc} is defined as the work of all the **nonconservative forces** of the system.